PHYS7101 · NUMERICAL METHODS AND DATA ANALYSIS

Exercise 1: The Large-Amplitude Pendulum

Weight	25% of the module	**Due**	4pm, Friday 30 October 2026, on Moodle
Covers	Weeks 1–3	**Submit**	Notebook (already run), ending with your statement (≤ 300 words, up to one figure) + GenAI declaration

1. The experiment

A pendulum (a compact bob on a light, rigid rod; treat it as a simple pendulum of length L = 0.800 m, taken as exact) was released from rest at about 70° from the vertical and filmed at 25 frames per second for 60 s. The angle θ from the vertical was extracted from every frame by video tracking. The tracking uncertainty is 0.5° per frame, independent from frame to frame. $t = 0$ is the frame of release.

Your job is to build a numerical model of the pendulum, show that you can trust it, use it to measure g, and decide how far you trust your answer.

For this exercise, we are using simulated data. Everyone has the same data.

2. Data and starting notebook

Everything is in the module repository, folder coursework/exercise1:

- **exercise1_pendulum.ipynb**: loads the data, and gives a reference solver (Task 2) and a model skeleton (Task 3). Open it in Colab, then **File → Save a copy in Drive**.
- **datasets/pendulum_angle.csv**: 1500 rows, one per frame (frame interval 0.04 s); columns `t_s` (time in s) and `theta_deg` (angle in degrees). Work in radians in your code; use degrees only for plots.

3. Required tasks

Task 1: Look at the data and write the model (Lecture 2)

Plot θ against time for the whole record, and for the first and last 10 s. Use the plot to make a first estimate of the period, and compare it with the small-angle period $T_0 = 2\pi\sqrt{L/g}$.

Start from the equation of motion with linear damping:

$$\ddot{\theta} + \gamma\,\dot{\theta} + \frac{g}{L}\sin\theta = 0.$$

Write it as two first-order equations for θ and $\omega = \mathrm{d}\theta/\mathrm{d}t$. Classify the system: linear or nonlinear, autonomous or non-autonomous, deterministic or stochastic, conservative or dissipative. Say what behaviour you expect, and what is conserved when $\gamma = 0$.

Report: the plots; your period estimate and T_0; the two first-order equations; the classification with reasons; the expected behaviour and the conserved quantity.

Task 2: Your own solver, and why you can trust it (Lectures 1 and 2)

Write Euler and RK4 functions for $\mathrm{d}\mathbf{y}/\mathrm{d}t = \mathbf{f}(\mathbf{y})$, building on your Week 1 lab RK4. Test them on the undamped pendulum ($\gamma = 0$), released from rest at 70°:

2.1 Energy. Plot the relative error of the energy per unit mass, $E = 1/2\,L^2\omega^2 + gL(1 - \cos\theta)$, over 10 s, for h = 0.04 s (the frame interval) and h = 0.004 s. What does Euler do to the energy, and why?

2.2 Order. Against the notebook's high-accuracy reference, find the largest error in θ over the first 2 s for h = 0.02, 0.01, 0.005 and 0.0025 s. From how it falls as h halves, find the order of each method.

2.3 Step. Choose your step h for Task 3 onwards (0.04 s must be a whole number of steps). Justify it with the $h/2$ test: compare the change in θ when h is halved with the 0.5° uncertainty.

Report: the energy plot and your explanation; a table of h, errors and error ratios, and the order of each method; your h and its $h/2$ difference compared with the uncertainty.

Task 3: Measure g by fitting (Lecture 3)

3.1 Small-angle fit. Fit the small-angle solution $\theta = A\,e^{-\beta t}\cos(\Omega t + \varphi)$ with `curve_fit` and appropriate weights, and convert Ω to g.

3.2 Numerical fit. Call your RK4 solver (step from Task 2) inside the model function you give to `curve_fit`; the notebook has a skeleton, `model_theta`. The free parameters are θ_0, g and γ. Test on the first 10 s, then fit the whole record.

3.3 Assess. Plot both fits with their normalised residuals underneath. What do the residuals and χ^2/ν tell you about each model? Compare your g, in units of your uncertainty, with the local value at Lancaster (54.0° N, near sea level) from a source you cite; not 9.81 or 9.80665 m/s².

Report: g and χ^2/ν from 3.1; θ_0, g and γ with their uncertainties, and χ^2/ν (your own function), from 3.2; the figure and your explanation; the local g with its source, and your difference from it in units of your uncertainty.

Task 4: Period against amplitude (Lectures 3 and 4)

4.1 Zero crossings. Find the times at which θ crosses zero upwards: locate each sign change, interpolate between the frames (e.g. a cubic spline), and find the crossing with your own bisection or Newton–Raphson function. Cross-check with `scipy.optimize.brentq`. Successive crossings give the periods.

4.2 Plot. Estimate the amplitude during each period (e.g. half the range of θ within it), and plot period against amplitude.

4.3 Exact period. For an undamped pendulum of amplitude θ_a,

$$T(\theta_a) = 4\sqrt{\frac{L}{g}}\;K(k), \qquad K(k) = \int_0^{\pi/2} \frac{\mathrm{d}\varphi}{\sqrt{1 - k^2\sin^2\varphi}}, \qquad k = \sin\frac{\theta_a}{2}.$$

Evaluate K with `ellipk` from `scipy.special` (careful: `ellipk(m)` takes $m = k^2$), and check one value against your own Simpson or Gauss rule. Draw $T(\theta_a)$ and the small-angle period on your plot.

4.4 Interpret. Is the pendulum linear? How do you know? Why do the periods measured at small amplitude scatter more?

Report: the number of periods; the iterations per crossing and the largest difference from `brentq`; your check of K; the plot with both theoretical curves; your answers to 4.4.

Task 5: Angular velocity and extrapolation (Lecture 4)

5.1 Velocity. Estimate $\omega = \mathrm{d}\theta/\mathrm{d}t$ from the data by central differences, and compare it with ω from your fitted model. Is the scatter what you expect from independent errors of 0.5° per frame?

5.2 Extrapolation. Using your fitted model, when will the amplitude fall below 1°? How far do you trust this prediction?

Report: the rms difference between the two estimates of ω, and the value you expect from the noise; the time at which the amplitude falls below 1°, and how far you trust it.

4. Extension (Task 6): what comes next?

Look at what Tasks 1–5 have shown you. Decide what the most important open question is, carry out the analysis that answers it, and check that analysis. In the notebook, say **why** you chose it: the reason should come from your results.

You are assessed on how well your choice follows from your results, and on how well it is done and checked, **not** on how many things you try. One well-chosen, well-tested analysis is worth more than several superficial ones.

Some questions that might help (not a list to work through):

- Are your residuals random, or is there a pattern in time? Averaging them in blocks helps.
- Does a fitted parameter change if you fit only part of the record?

- Is the model complete? What else might slow a pendulum down?
- Is your quoted uncertainty believable? (A χ^2 scan, Week 2; or test your method on simulated data where you know the answer.)
- Does your result depend on the solver step, or on the solver? (Week 1)
- Would a moving-window polynomial fit for ω, or checking that the energy lost equals $\int \gamma L^2 \omega^2 \, \mathrm{d}t$, tell you something new?

5. Interpretive statement (≤ 300 words, up to one figure)

Write a short statement, in your own words, that answers: What is your best estimate of g, with its uncertainty, and how did you reach it? Is the pendulum linear, and which model describes it? Why can you trust your numerical solver? What did your extension show, and how confident are you now? What would you do next?

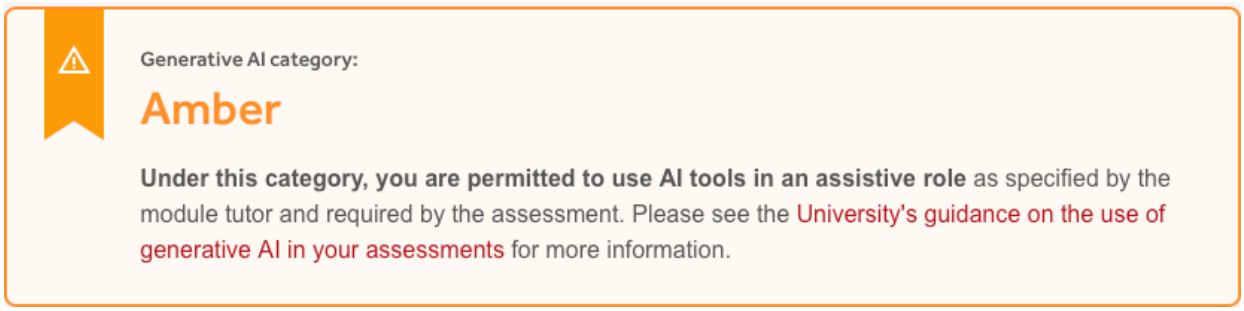
Base every claim on a result in your notebook, and quote the numbers. Put the statement at the end of your notebook, under the heading **Interpretive statement**, with a word count. It may include **one** figure that carries your key result; an A+ needs one. If you prefer, submit the statement and its figure as a separate PDF instead, and say so at the end of the notebook.

6. Tips:

- Before submitting ensure that your notebook runs top to bottom with no edits: **Runtime → Restart session and run all**.
- Before each block of code, include a short Markdown cell saying what you are doing and why; after it, what the result means.
- Write functions for anything you do more than once (your solver, your right-hand side $\mathbf{f}(\mathbf{y})$, your root-finder, your χ^2/ν).
- Figures should include: axis labels with units, legends, uncertainties shown, log axes where they help. See `week2_code_and_plots.ipynb`.

7. Submission Guidance, GenAI and help

- Submit on Moodle by 4pm, Friday 30 October 2026:
  - the **.ipynb notebook, already run**, with all outputs visible and your **statement** at the end (or as a separate PDF)
  - the **GenAI declaration** (mandatory). For an extension to the deadline, use the request form on the Part II Moodle page.
- GenAI:



  - may be used to assist with code development, debugging and to explore analysis ideas. You must be able to explain every line you submit. Your statement, and the reasons for your choices in the notebook, must be your own words; GenAI may only correct spelling and grammar.
- Do ask questions in the week 3 lab.
- The feedback session will take place at 13:00 on Friday 6 November 2026, where we go through the solutions and discuss common issues. We will aim to return marked submissions as soon as possible after the deadline.

8. Suggested plan (about 12 hours in total)

When	Suggested work	Time
Before the Week 3 lab	Tasks 1–3, using your RK4 solver from the Week 1 lab and your fitting code from the Week 2 lab.	5–6 h
Week 3 lecture and lab	Tasks 4 and 5 (root-finding, interpolation, quadrature, finite differences). Then decide on your extension from your Task 1–5 results; discuss it with a PGTA.	2–3 h + lab time
Week 3 lab to the deadline	Carry out and check the extension; write the statement; restart and run all; submit.	3–4 h

Marking grid

NB: At level 7, D and below is a fail. A+ is reserved for work where no improvement can be suggested.

University scale	Results (Tasks 1–5) 50%	Extension (Task 6) 15%	Interpretive statement 15%	Coding 12%	Visualisation 8%
Exceptional (1st Class) A+ [24]	Everything in Excellent, with every result confirmed by an independent check. Precision and uncertainties exactly as the evidence supports.	Goes beyond the obvious next step: finds and quantifies a subtler effect, with independent checks that agree. Could be handed to another scientist as it stands.	Concise and compelling, with one figure that carries the key result. Every claim quantified. Separates statistical from systematic uncertainty, states the limits of the conclusion and the most valuable next measurement.	Clean, reusable functions, tested on cases with known answers. Another student could reuse it without asking.	Figures that make the key result obvious at a glance, each designed for the point it supports.
Excellent (1st Class) A− A [18, 21]	All required results correct, with uncertainties, units and sensible precision. Methods justified and checked.	Clearly motivated by your own results. Appropriate, correctly done and checked independently. Outcome quantified and used in your conclusions.	A best estimate with its uncertainty. Explains, with your numbers, why results agree or differ. Judges how far to trust them. A specific next step.	Runs top to bottom. Functions for repeated work. Markdown explains why. Readable.	Clear, labelled with units, uncertainties shown, residuals under the fit, each figure used in the argument.
Good (Upper 2nd) B− B B+ [15, 16, 17]	Results correct, with minor slips. Limited justification or checking.	A sensible next step linked to your results, done correctly. Limited checking.	Mostly supported by results; describes more than it explains. Next step generic.	Runs. Mostly clear; some repetition or thin explanation.	All figures present and labelled; one weakness.
Satisfactory (Lower 2nd) C− C C+ [12, 13, 14]	Partly correct, or a method error. Uncertainties missing or misused.	Only loosely linked to your results, or partly incorrect. Outcome not quantified.	Mainly a description of what was done. Little explanation. Numbers without uncertainties.	Runs only after small fixes, or hard to follow.	Figures hard to read, or key elements missing.
Weak (3rd Class) D− D D+ [9, 10, 11]	Largely incorrect or incomplete. No uncertainties.	A token analysis with no stated reason, or wrong without awareness.	Conclusions not supported by, or contradicting, your results.	Does not run, or results cannot be traced to code.	Unlabelled or misleading figures.
Fail F4 F3 F2 F1 [0, 2, 4, 7]	Little or nothing usable.	Absent, or trivial.	Missing, irrelevant or largely incorrect.	No usable code.	No figures.